CSC4010: Introduction to Theoretical Computer Science
Problem Set 7

Overview

One of the goals of this homework will be to practice showing that certain problems are NP-hard. Towards this, it might be useful to have a (non-exhaustive) list of well-known NP-hard problems to keep in mind:

1. **Vertex Cover (VC)**

   Input: A graph $G = (V, E)$ and an integer k.

   Question: Does there exist a set $C \subseteq V$ with $|C| \leq k$ such that every edge has at least one endpoint in C? Equivalently, $\forall \{u, v\} \in E,\ u \in C$ or $v \in C$.

2. **Independent Set (IS)**

   Input: A graph $G = (V, E)$ and an integer k.

   Question: Does there exist a set $I \subseteq V$ with $|I| \geq k$ such that no two distinct vertices in I are adjacent? Equivalently, $\forall u, v \in I, \quad u \neq v \implies \{u, v\} \notin E$.

3. **Set Cover (SC)**

   Input: A finite universe U, subsets $S_1, \ldots, S_m \subseteq U$, and an integer k.

   Question: Can at most k of the subsets cover all of U? Equivalently, does there exist $J \subseteq \{1, \ldots, m\}$ such that $|J| \leq k$ and $\bigcup_{j \in J} S_j = U$?

4. **Hamiltonian Path (HP)**

   Input: A graph $G = (V, E)$ and distinct vertices $s, t \in V$.

   Question: Is there a path from s to t that visits every vertex exactly once?

5. **Hamiltonian Cycle (HC)**

   Input: A graph $G = (V, E)$.

   Question: Is there a cycle that visits every vertex exactly once before returning to its starting vertex?

6. **3-Satisfiability (3SAT)**.

   Input: A Boolean formula F in conjunctive normal form, with at most three literals in each clause.

   Question: Is there a truth assignment satisfying every clause of φ?

7. **Clique**.

   Input: A graph $G = (V, E)$ and an integer k.

   Question: Does there exist a set $C \subseteq V$ with $|C| \geq k$ such that every pair of distinct vertices in C is adjacent?

8. **3-Coloring**.

   Input: A graph $G = (V, E)$.

   Question: Is there a coloring $c : V \to \{1, 2, 3\}$ such that $\forall \{u, v\} \in E,\ c(u) \neq c(v)$?

9. **Subset-Sum**

   Input: Nonnegative integers $a_1, \ldots, a_n$ and a target T.

   Question: Does there exist a subset of indices $J \subseteq \{1, \ldots, n\}$ such that $\sum_{j \in J} a_j = T$?

Questions

1. (Search-to-Decision) Suppose you have access to a **SAT decision oracle** that given any Boolean formula $F(x_1, \cdots, x_n)$, returns `YES` if F is satisfiable and `NO` otherwise. The oracle does *not* return a satisfying assignment. Design a deterministic *polynomial-time algorithm* that solves this search problem using at most $n + 1$ calls to the SAT decision oracle. That is, your algorithm should output a satisfying assignment iff F is satisfiable.

2. (CLIQUE is NP-hard) In this question we will prove that CLIQUE is NP-hard by constructing a polynomial-time mapping reduction $3\text{SAT} \leq_p \text{CLIQUE}$.

   Let the input formula be $\varphi = C_1 \wedge C_2 \wedge \cdots \wedge C_m$, where $C_i = (\ell_{i,1} \vee \ell_{i,2} \vee \ell_{i,3})$, and $m \geq 1$ and each literal $\ell_{i,j}$ is a variable or the negation of a variable.

   (a) Construct the graph G_φ corresponding to φ as follows. For each literal occurrance $\ell_{i,j}$ create a vertex $v_{i,j}$. Give a precise rule for adding an edge between two distinct vertices $v_{i,j}$ and $v_{r,s}$. Specify the threshold k in terms of m, and explain your choice.

   (b) What is the time taken to construct G_φ given φ? Can you bound the size of G_φ (i.e. the number of vertices and edges) in terms of φ.

   (c) Show that if φ is satisfiable then G_φ contains a clique of size at least k.

   (d) Show that if G_φ contains a clique of size at least k then φ is satisfiable.

   (e) From the preceding parts, conclude that CLIQUE is NP-hard.

3. Consider the problem Double-SAT$= \{F \mid F$ is a Boolean formula that has at least two satisfying assignments$\}$. Show that SAT $\leq_p$ Double-SAT.

4. Show that SAT $\leq_p$ 3-SAT.

5. The complexity class coNP is defined as $\text{coNP} = \{L \mid \overline{L} \in \text{NP}\}$. That is, coNP is the set of languages whose complements are in NP. (Note that this is not the set of languages that are outside NP!) Suppose $L_1, L_2 \in \text{NP} \cap \text{coNP}$. Show that $L_1 \oplus L_2 \in \text{NP} \cap \text{coNP}$, where $L_1 \oplus L_2 = \{x \mid x$ is in exactly one of L_1 and $L_2\}$.